

Network Technical Report

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IFT 372: Wireless Fundamentals

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Abstract

This technical report provides a thorough overview and showcase of a functional network including wired and wireless components. It uses mechanisms that serve to support public and private routing, IP address space reusability, network security, data isolation, and end-user services. A few mechanisms used are Dynamic Host Configuration Protocol (DHCP), Network Address Translation (NAT), Enhanced Interior Gateway Routing Protocol (EIGRP), Voice over IP (VoIP), and WiFi Protected Access - Preshared Key (WPA2-PSK). The network topology interconnects 3 Local Area Networks (LANs) so that each may communicate with the other and access shared resources existing outside of them. This network is also designed to have access to the internet instead of just being an air-gapped network, the appropriate configurations were made to direct public traffic outside of the network. The final network is flexible, scalable, well-performing, and secure.

Keywords: Networking, IP routing, Wireless Technology, Telephony, LTE

Introduction

This assignment called for me to build a network showcasing various wireless and general networking concepts and implementations. Specifically, we were called to create three separate Local Area Networks (LANs) and connect them so that they can communicate with each other via voice and data. A practical way to look at this assignment is to say you are an ISP creating a network for customers to use. You would want your network to have as much reach as possible. the network should support as many technologies for your customers to choose from. Customers can choose wired plans that supply internet to their home network like a typical household LAN.

You can also sell to customers who want cellular connectivity over 4G/5G LTE. You should also make sure your network supports the services that your customers need such as voice and data. The goal of this network is to provide a variety of wired and wireless technologies and have reliable reachability between other local networks and the Internet.

Design

This assignment requires you to have three distinct LANs. LAN 1 is a cellular network. It must contain devices such as cell towers, smartphones, and a central office server to showcase a wireless network that is based on LTE. LAN 2 is a WiFi network, similar to what you would have in your home or office. It must contain devices such as access points, laptops, smartphones, and a switch. LAN 3 is a mixed network. It must contain devices such as computers, IP phones, a server, and a switch. Each of these LANs must have a gateway router for inbound and outbound traffic. There also needs to be security controls implemented in the networks, security measures can be specific to the type of network.

Development

Figure 1.1

Overall Network Topology

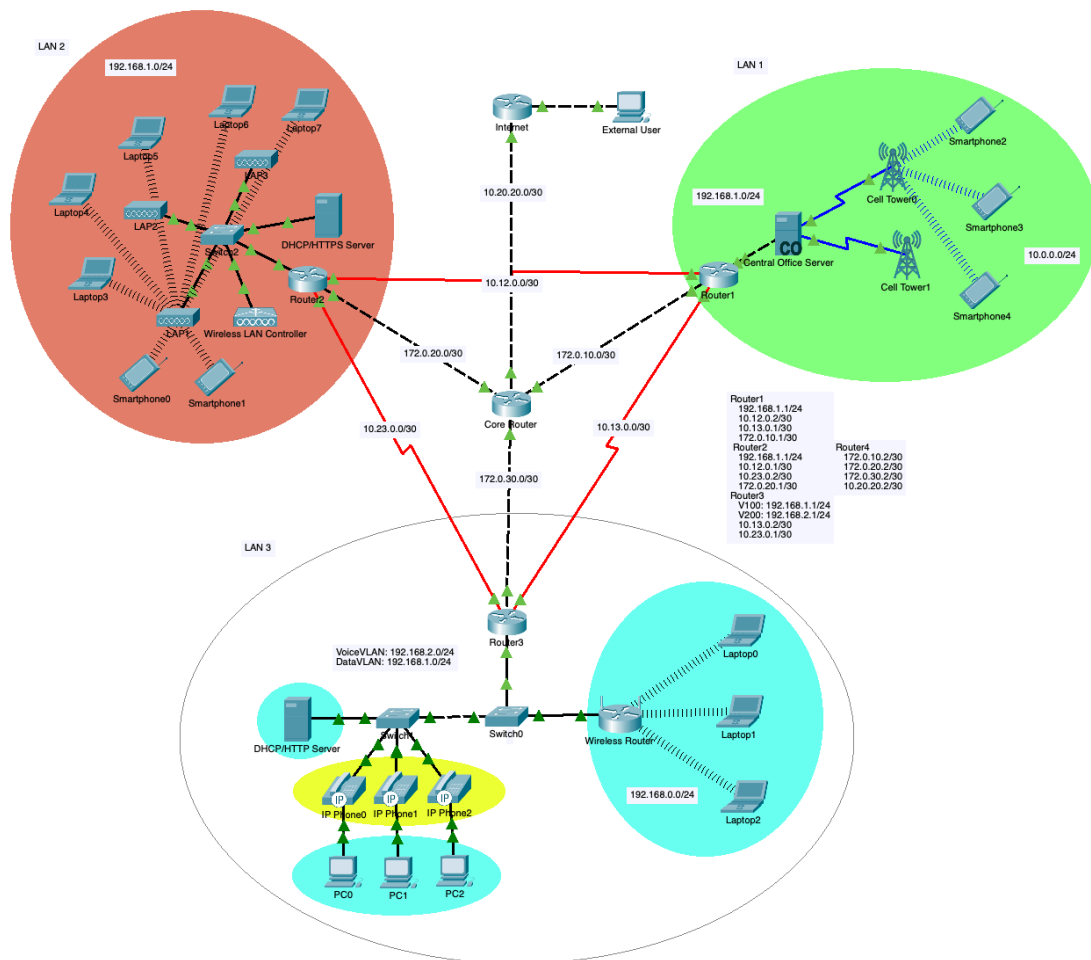


Figure 1.2

LAN 1: Cellular network.

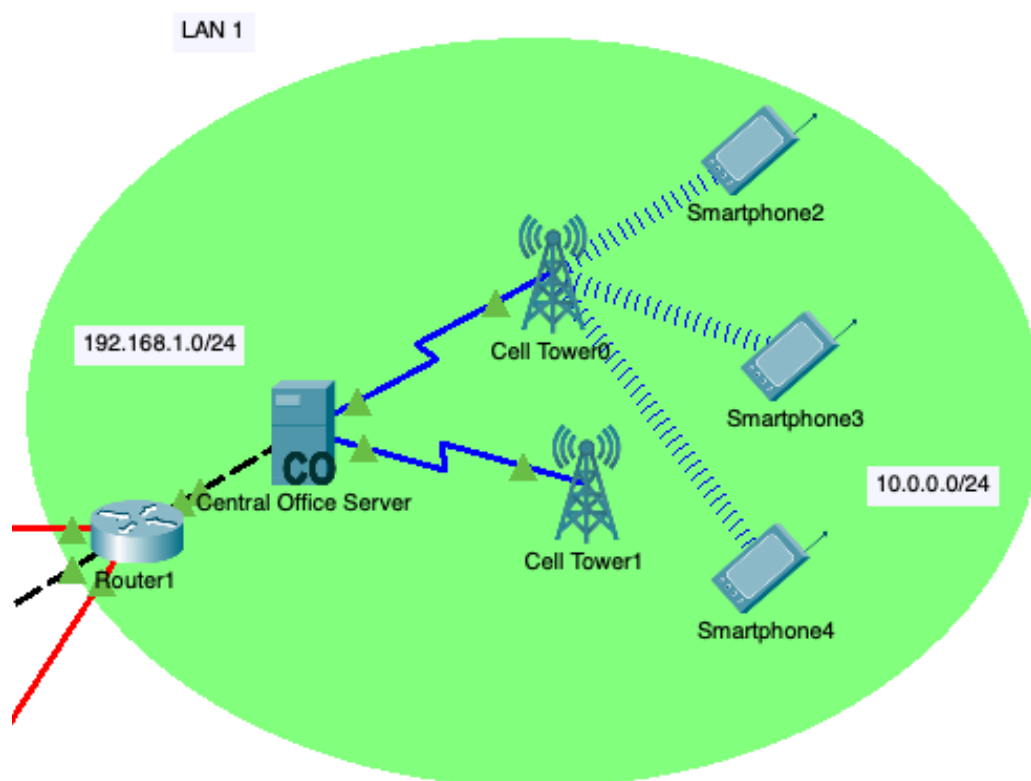


Figure 1.3

LAN 2: WiFi network.

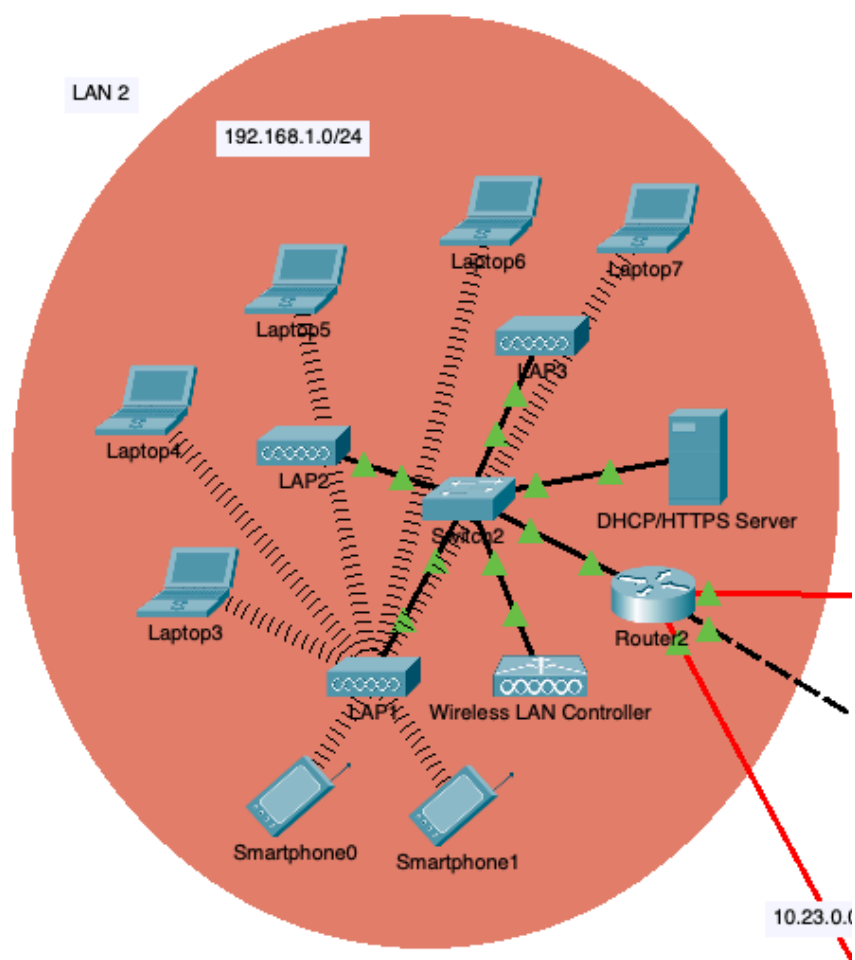


Figure 1.4

LAN 3: Mixed network.

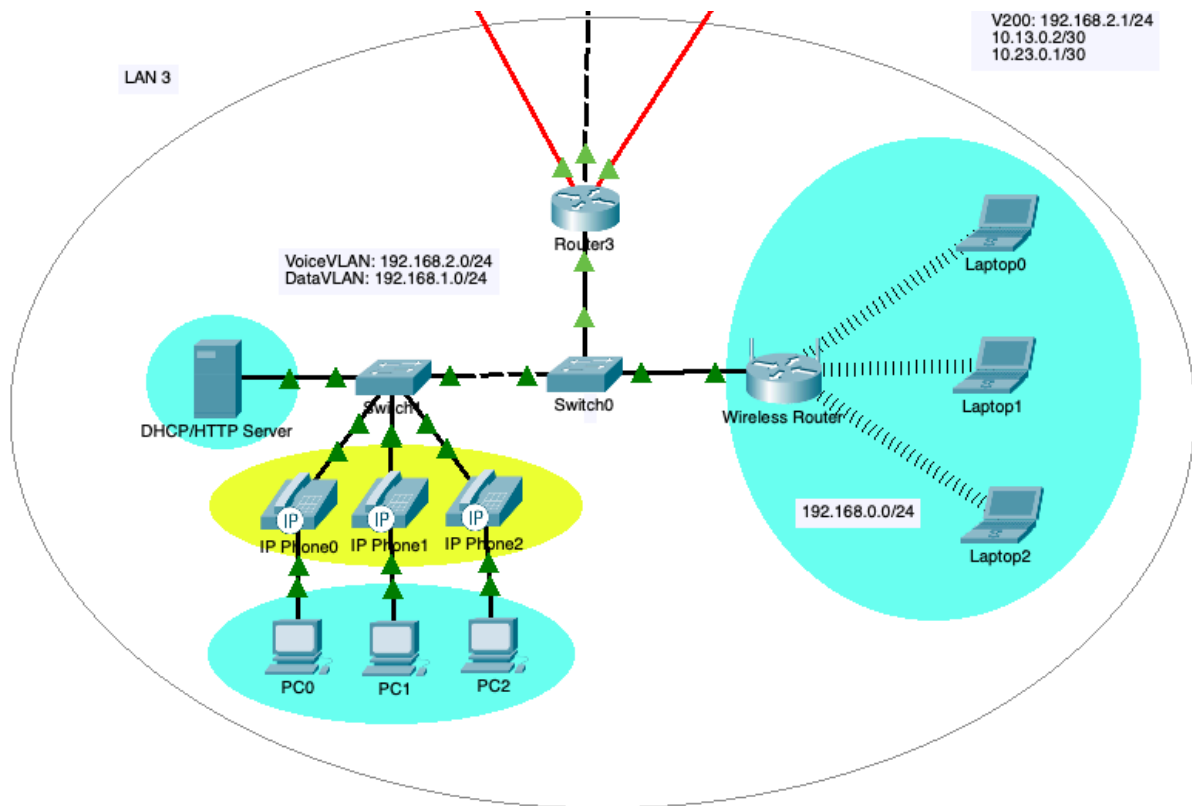
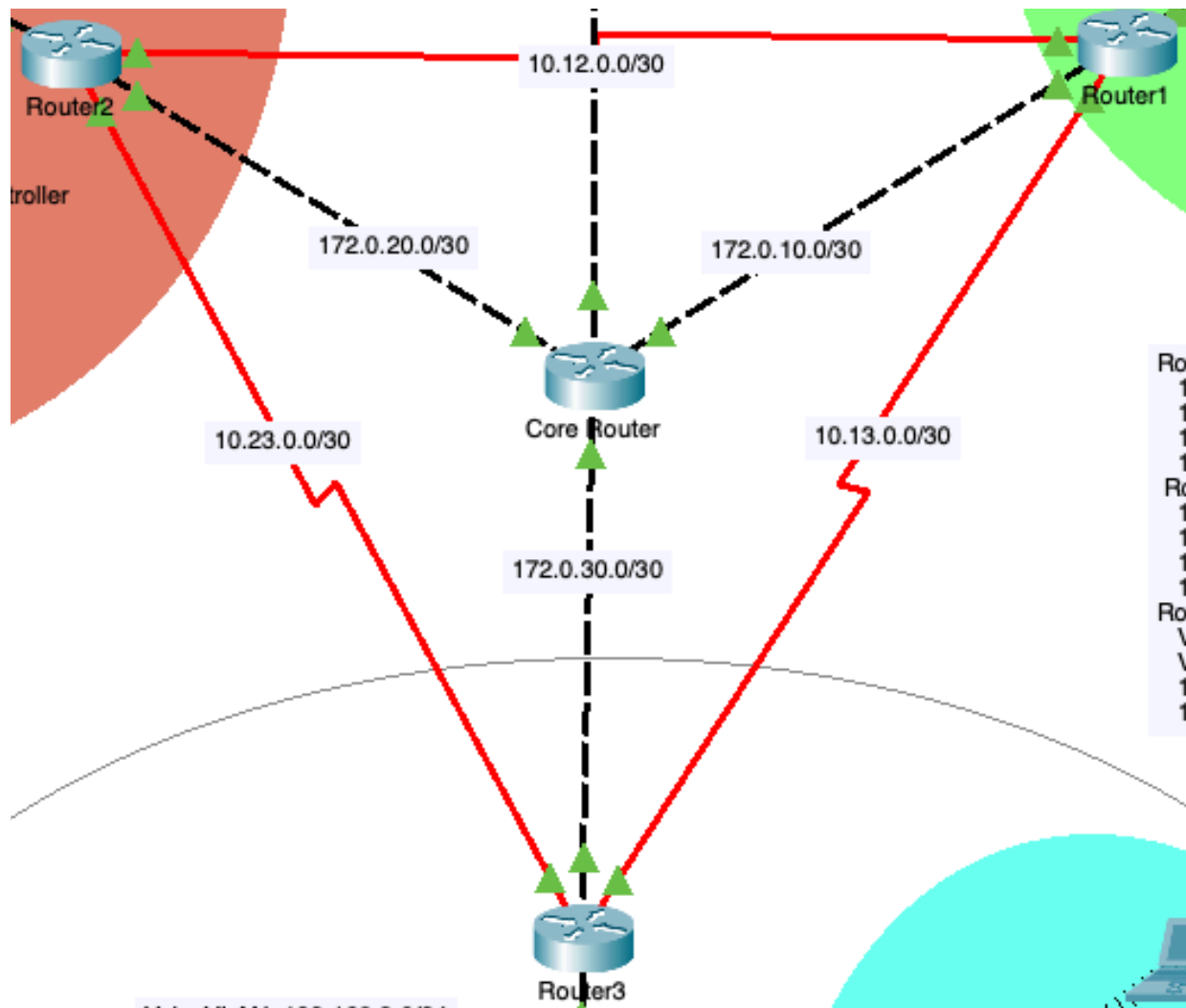


Figure 1.5*Central networks.*

As you can see in the above figures, I have at least met the visual requirements. LAN 1, 2, and three have the required components and work as intended on their own. Hosts get IP addresses and other fields via DHCP and are all reachable within their respective LANs. In LAN 1, The smartphones connect to the cell towers via LTE. The cell towers are connected to the local network via the Central Office Server. The COS provides DHCP for the smartphones and serves

as the default gateway for them as well. The COS connects to the gateway router via its backbone interface under a different subnet than the cell tower interface that talks to the smartphones. In this respect, the COS is similar to a router in that it separates distinct networks and routes packets between them. In LAN 2, there is a switch connecting to three Lightweight Access Points, a Wireless LAN Controller, and a server. The LAPs are connected to laptops and a couple of smartphones. I could have made a simpler setup by using regular Access Points and configuring them all to have the same SSID, authentication method, and password but I wanted to find a more elegant solution. I came across the Wireless LAN Controller in my research and decided it was the right solution for this LAN. It allows me to define one or more WLANs and set the SSID, authentication method, and password centrally for the LAPs to get their configuration from. All of the configuration is done on the WLC so I do not need to configure each AP. Another advantage of using the WLC is that it has a DHCP server installed out of the box. Although using the WLC's DHCP server is not required, I still chose to use it over the standalone server on the network. This was my decision, I could disable the WLC DHCP server and use the standalone server's DHCP and it'll work just fine. This method of central configuration for WLANs is very scaleable, flexible, and convenient, which is perfect for LAN 2 as it requires many APs for users and it would be a pain to update the configuration on every single AP (Smith et al., 2011). This design allows for connected wireless devices to roam across the LAPs depending on the signal strength of each. This is something that is difficult to demonstrate in Cisco Packet Tracer where spacial reference is lacking but this can be observed in reality. The central switch is connected to the gateway router which has IP telephony configured with Call Manager Express. In LAN 3, there are two switches connected via a trunk for VLANs, a server, a wireless router, three laptops, and three IP phones connected to three PCs in series.

This LAN is split into three networks; 192.168.0.0/24, 192.168.1.0/24, and 192.168.2.0/24. The 0.x network is provided by the wireless router and is occupied by the three laptops. The 1.x network is occupied by the server, PCs, and wireless router's internet interface. The 2.x network is occupied by the IP phones. The gateway router's LAN interface is split into 2 subinterfaces for VLANs 100 and 200 which belong to 192.168.1.0/24, and 192.168.2.0/24 respectively. It also has Call Manager Express setup for the IP phones and is configured for the IP phones to query the server for DHCP instead of the gateway router. Implementing VLANs allows for the IP phone voice traffic to be isolated from the rest of the network for security and QoS purposes. Certain software exists to manage data streams such as those of VoIP for optimizing QoS for those services. NetEqualizer is a popular product that was designed for this ("NETEQUALIZER UNVEILS", 2007). This LAN implements a wireless router behind the gateway router, which performs NAT similarly to the wireless router. This setup may cause performance problems in certain situations due to double NAT. With double NAT address translation is performed twice unnecessarily. This is essentially unnoticeable with light traffic but the slight loss in performance adds up when traffic is high. The wireless router also serves DHCP to the laptops on its inner LAN. Each of the three gateway routers performs NAT to translate the private addresses of the LANs to the private and public addresses used in the central networks. This allows for IP address space reuse and added security. All three LANs use the same IP address block of 192.168.1.0/24. This is only possible because of NAT since the source IP address of outbound LAN traffic is changed to the outside IP address rather than the actual device's private IP address that initiated the traffic. NAT makes the router act like a messenger that makes the connections on its own behalf and returns the data to the initiating host when it gets a response from the destination. This adds a layer of security for the LANs because external hosts cannot make direct connections to

the hosts behind NAT, the LAN host must initiate the connection for any type of communication. This makes peer-to-peer communication difficult. P2P networking solutions such as ZeroTier leverage NAT traversal techniques like UDP hole-punching to achieve P2P communication (Ford et al., 2006). Each gateway router has an inside interface and three outside interfaces with the exception being Router3 which has two inside subinterfaces. I used Port Address Translation (PAT) rather than Static NAT because I find it easier to configure than Static NAT for this network. Each gateway router has 2 serial connections going to the other gateway routers for direct, fast communication to the other LANs and a single FastEthernet connection with the public IP addresses of 172.0.10.1, 172.0.20.1, 172.0.30.1 for the respective LANs so that they are publicly routable on the internet. Each of these public interfaces is connected to a core router that leads out of the network. I have set the gateway of last resort on each gateway router and the core router to point traffic that is destined for addresses on the internet outside of the network instead of being dropped by the routers because those addresses are not in the routing tables. Each router is configured with EIGRP to handle routing. This network is much too complex to easily set up static routes for each router. EIGRP updates the tables on changes in the networks automatically.

Testing and Results

I will test each LAN according to the requirements for each. I will test for data and voice and prove network reachability for each.

Figure 2.1

Pinging LAN 2 from LAN 1. Starting point.

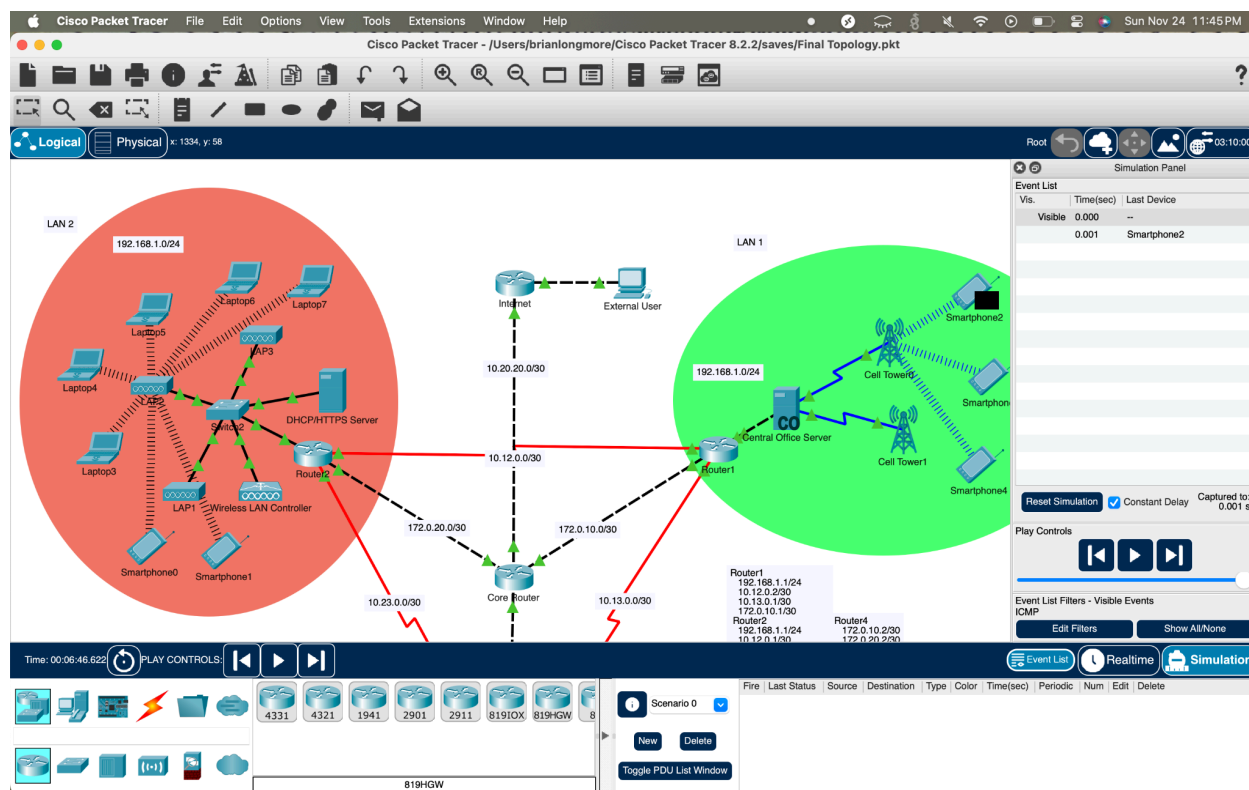
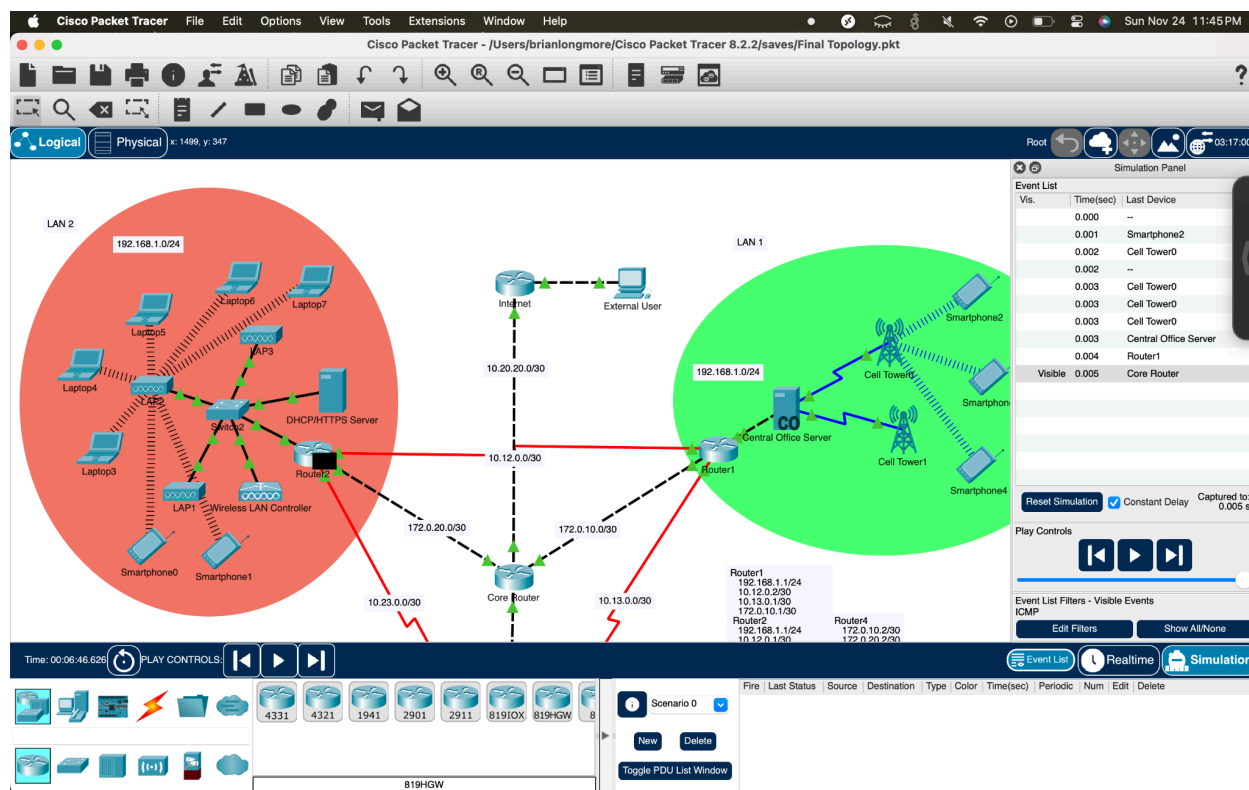


Figure 2.2

Pinging LAN 2 from LAN 1. End point.



Note: This demonstrates network connectivity between LAN 1 and LAN 2.

Figure 3.1

Pinging LAN 3 from LAN 2. Starting point.

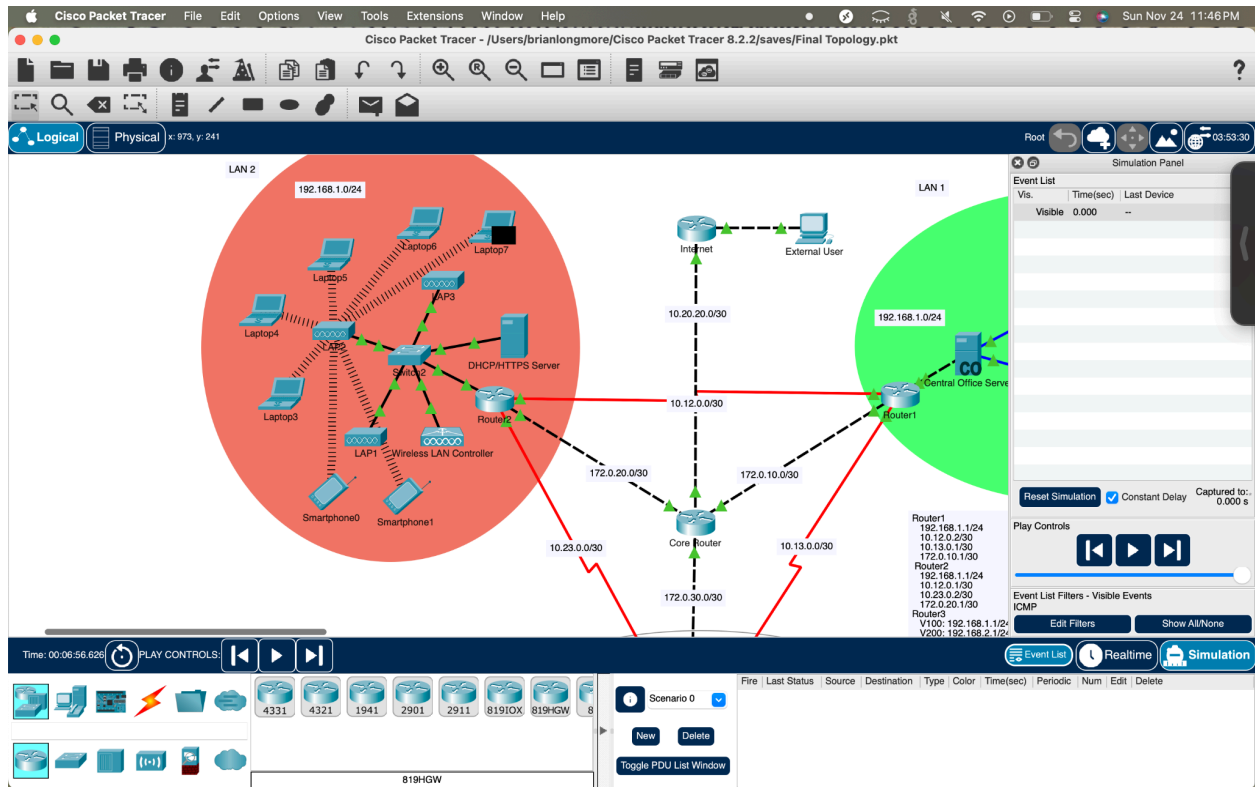
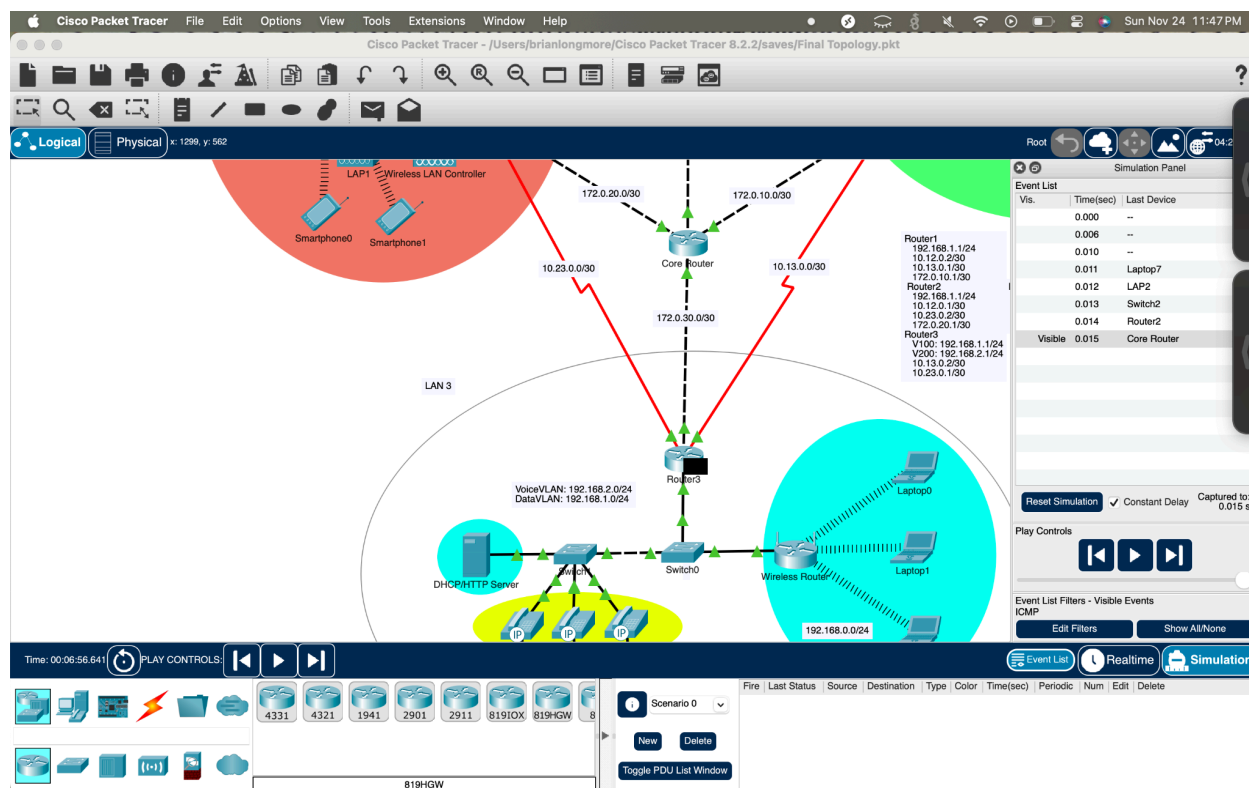


Figure 3.2

Pinging LAN 3 from LAN 2. End point.



Note: This demonstrates network connectivity between LAN 2 and LAN 3.

Figure 4.1

Pinging LAN 1 from LAN 3. Starting point.

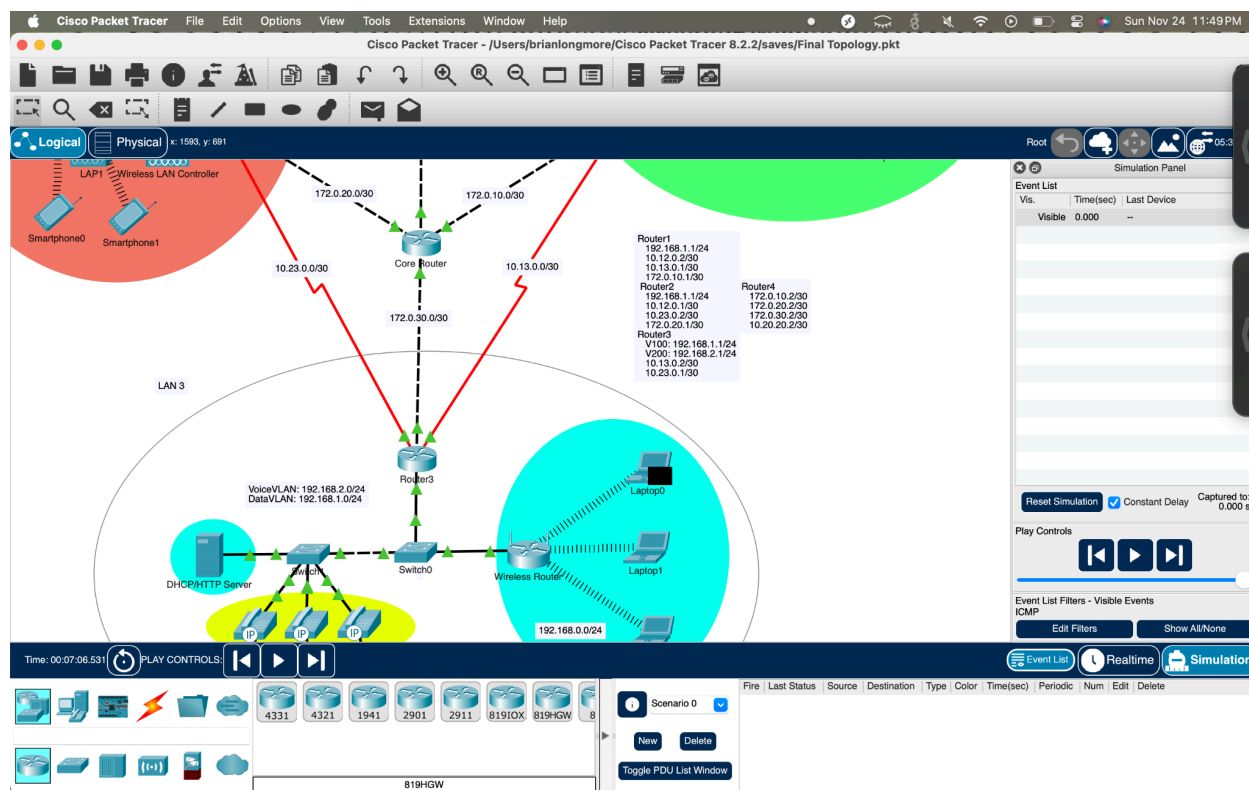
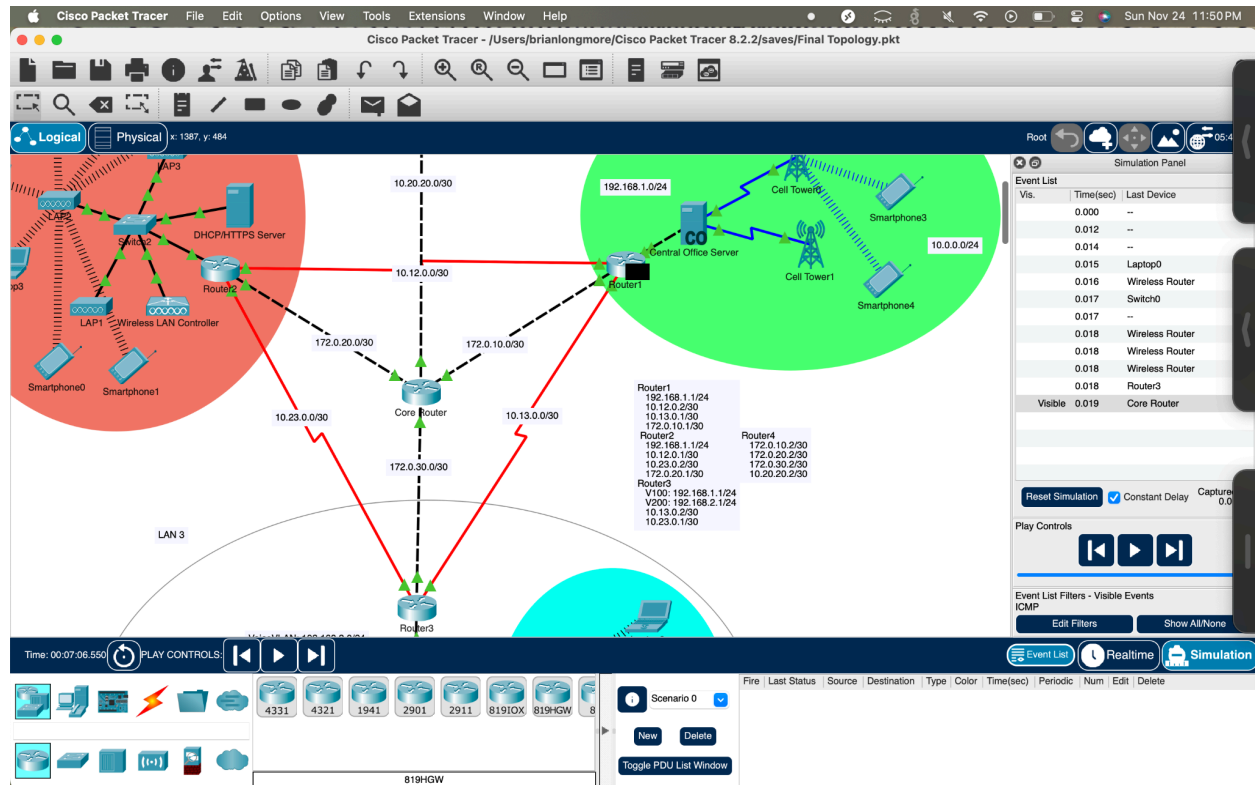


Figure 4.2

Pinging LAN 1 from LAN 3. End point.



Note: This demonstrates network connectivity between LAN 1 and LAN 3.

Figure 5.1

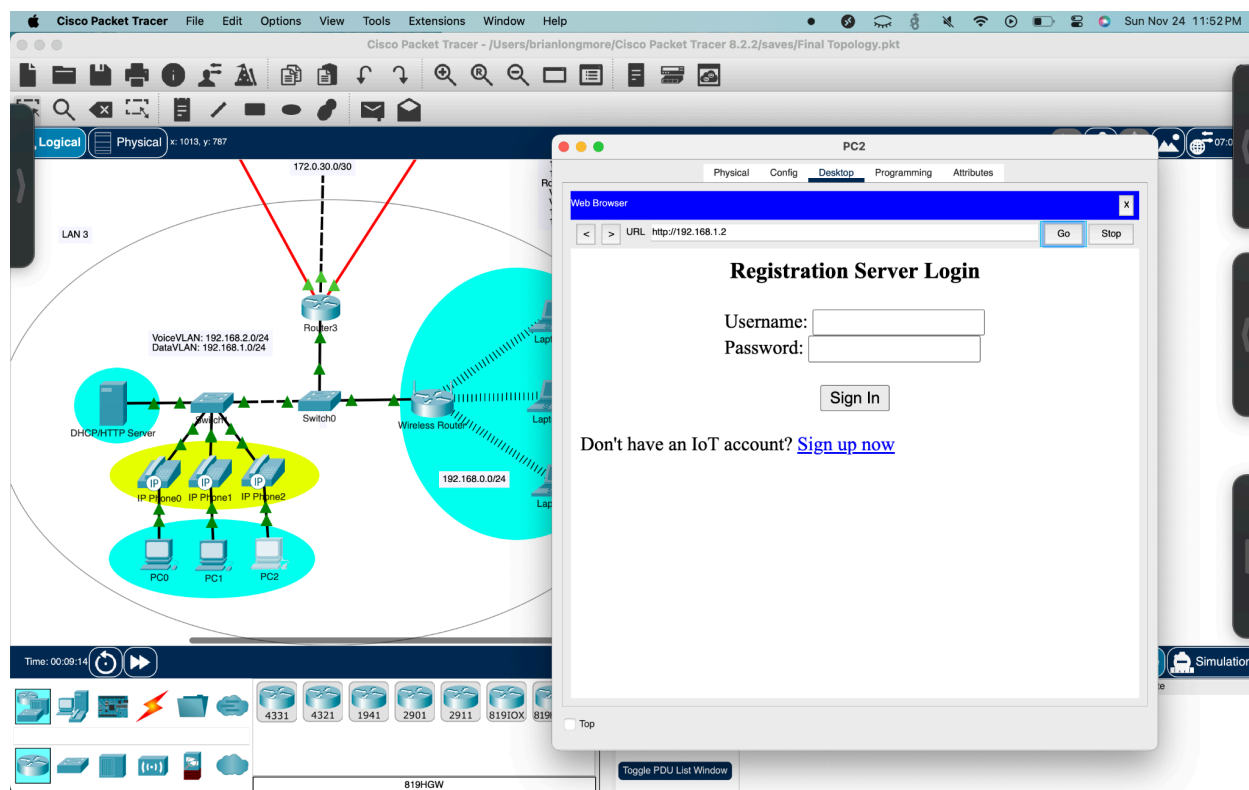
Starting a voice call in LAN 3.



Note: This demonstrates IP telephony capability in LAN 3.

Figure 5.2

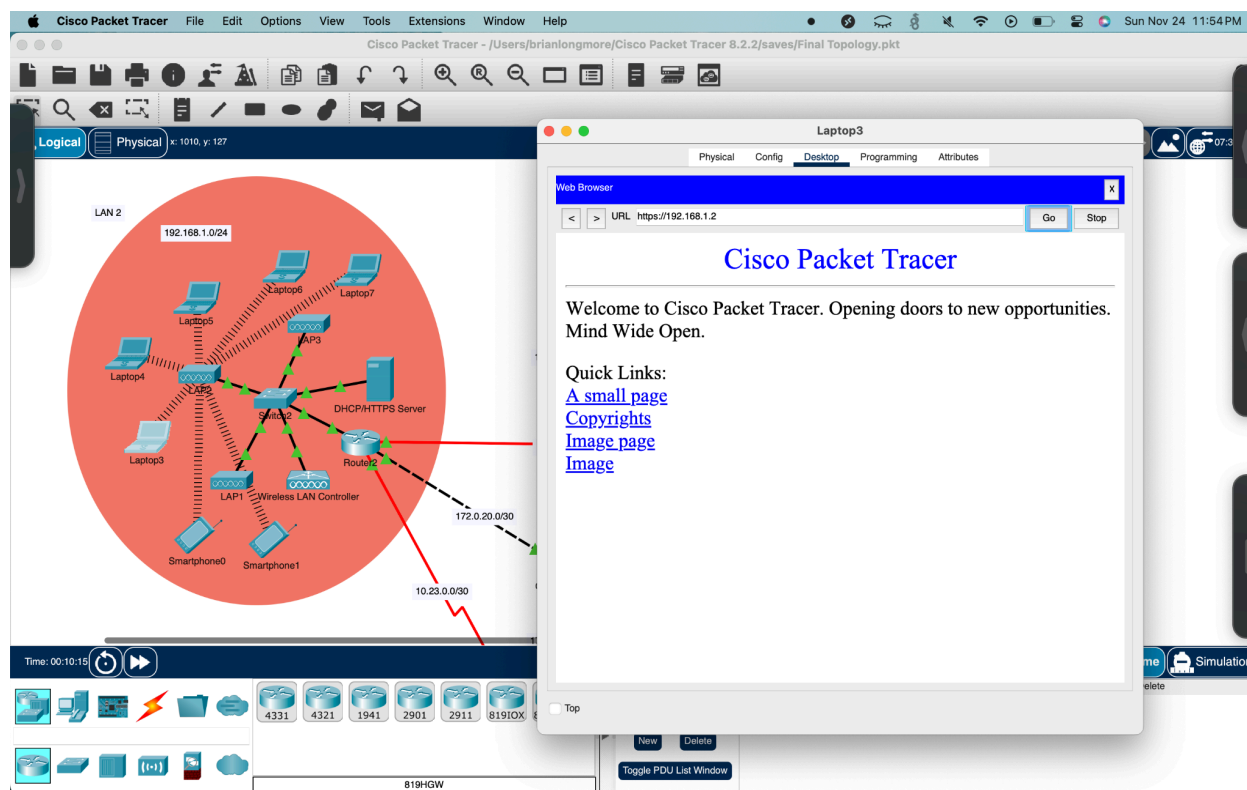
Showcasing a Client-Server exchange process in LAN 3 with HTTP server.



Note: This demonstrates intranetwork reachability in LAN 3.

Figure 5.3

Showcasing a Client-Server exchange process in LAN 2 with HTTPS server.



Note: This demonstrates intranetwork reachability in LAN 2.

Conclusion

This assignment taught me a lot about wireless architecture and general networking concepts. Many of the things I did in this assignment were the first time I've ever done them. For example, this was my first time configuring NAT to work with more than one inside interface and one outside interface. This was also my first time using EIGRP, the only dynamic routing protocol I've used before this was RIP. I was amazed by how much time and effort it saved over setting all the routes statically. A couple of times, the EIGRP output logs on the router CLIs started to show that there was a problem in my network. I realized that I had misconfigured an interface subnet

on a router and fixed it. It was cool using the EIGRP logs to know when I had done something incorrectly. I also learned a lot about using Packet Tracer, the certain buttons that you have to click, and where to find them. I got much more comfortable using the Cisco IOS CLI from building this topology and configuring the networking devices and am much more prepared to use the CLI again in the future. This learning took much self-study and hands-on experimentation to get things working.

References

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